# Supplementary Material
# *Physics Is Already There:* Rethinking Physical Injection in Latent World Models

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## A Training and Evaluation Details

**Reproducibility.** All training configurations, per-run logs, evaluation protocols, and the exact numbers behind every table and figure are archived alongside the code; the study builds on public assets (LeWM, PhyWorld, Physion, Physion++). We will release the evaluation suite, including the per-horizon/per-partition audit tools and the random-encoder controls.

**Hyperparameters.** All PhyWorld runs: AdamW, lr $5{\times}10^{-5}$ (from-scratch reruns: $2{\times}10^{-5}$), weight decay $10^{-3}$, bf16, gradient clip 1.0, 20 epochs (60 for from-scratch cells; the pretraining-regime $2{\times}2$ uses 60 epochs uniformly), batch size 64 for free rollout ($H$=8), 128 for teacher forcing ($H$=1), 48 for $H{\geq}16$. History size 3; embedding dimension 192; SIGReg weight 0.09. Seeds: 3072/1234/42 where seed-replicated. Physion++ runs use the readout split (800 clips) with the same optimizer.

**OOD partitions.** ID: radius/mass $\in [0.7, 1.5]$, velocity $\in [1.0, 4.0]$. OOD draws radius/mass from $[0.3, 0.6]{\cup}[1.5, 2.0]$ and velocity from $[0, 0.8]{\cup}[4.5, 6.0]$. Collision uses the four-column variant (two objects' masses and velocities). Training sets are ID-only (1,000 trajectories); evaluation covers all four partitions (${\geq}$1,600 trajectories per domain).

**Probe protocol.** Ridge regression ($\alpha$=1), trajectory-level 80/20 splits, probing the encoder output (not the projector), pretrained "paper" initialization, and $K$=4 stacked consecutive latents for velocity. Each choice matters: the naive combination (random init, single frame, through-projector) reads $\rho$=0.166 for uniform velocity where the corrected protocol reads 0.939 (Trap 4, Appendix E). Random-encoder and pixel-statistics controls are run alongside every probe.

## B Full Injection Results

Table 1 is the numeric form of Figure 2 of the main paper, with the seed annotations spelled out.

**Probe/slot factorial (uniform, free rollout).** Latent both-OOD nMSE / pixel both-OOD PSNR (dB): baseline 0.131 / 20.41; probe-only 0.167 / 19.83; slot+reweight 0.114 / 21.30; probe+slot+reweight 0.125 / 20.02. The combination underperforms the slot alone on both scales; the latent and pixel verdicts agree.

| | uniform (both-OOD) | parabola (r/m-OOD) | collision (both-OOD) |
|---|---|---|---|
| Free-rollout baseline | 0.131 | 0.127 | 0.393 |
| Fixed slot | 0.183 | 0.156 | 0.651 |
| +reweight ($w$=30) | 0.114[s] | 0.160 | 0.596 |
| +velocity | 0.207 | **0.096**[m] | 0.621 |
| Probe | 0.167 | 0.115[n] | 0.647 |
| +slot+reweight | 0.125[s] | 0.127 | 0.607 |
| Kinematic (free MLP) | 0.155 | 0.178 | 0.560 |
| Kinematic (strict $a$=$g$) | 0.206 | 0.173 | 0.559 |
| Consistency | 0.151 | 0.147 | 0.57–0.64 |
| Label-free prior | 0.171 | 0.172 | 0.653 |
| Grounded prior | 0.166 | 0.156 | 0.525 |

Table 1: The 30-cell scan (hardest clean split, latent nMSE ↓). [m]The one seed-robust gain: $0.096 \pm 0.007$ over three seeds vs. baseline $0.122 \pm 0.007$, non-overlapping. [n]Refuted by seeds: three-seed $0.137 \pm 0.034$ vs. baseline $0.122 \pm 0.007$—the single-seed low was seed luck. [s]Also refuted by seeds: reweighted slot $0.132 \pm 0.017$ and probe+slot $0.142 \pm 0.017$ vs. baseline $0.136 \pm 0.008$ (parity). Remaining cells are single-seed but sit 5–20 seed-std above baseline.

**Slot reweighting sweep (uniform).** Slot weight $1/30/100$: $0.183$ / $0.114$ / $0.136$ (single seed), with the $w$=30 cell seed-replicated at $0.132 \pm 0.017$ against the baseline $0.136 \pm 0.008$. Kinematic head on the reweighted slot: velocity-OOD decoded position $\rho$=0.991/0.987 ($x$/$y$); uniform pixel horizon-28 23.34 vs. 22.09 dB (+1.25); appearance-OOD remains the weak axis (decoded $\rho$ $0.29 \rightarrow 0.43$, vs. 0.96 without the kinematic head).

**Bypass probe details.** Protocol for §4.3 of the main paper, which reports the numbers. Two models are probed: the free-rollout baseline (no injection) and the slot-pinned model (structpos, $w$=30). For each, a ridge probe is fit on the dimension subset alone—the full 192, the black box [2:192], the slot [0:2], or a random pair as control—with train/test trajectories split 80/20, so decodability is measured out of sample. Values are Pearson $\rho$ against ground-truth position on the both-OOD partition, averaged over both coordinates, from frame embeddings rather than rolled-out latents. The slot absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$)

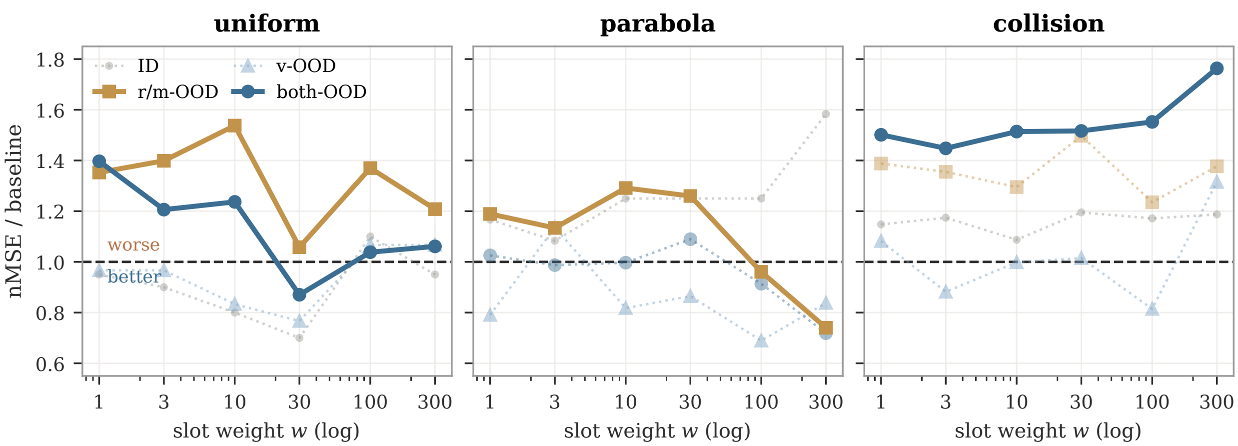


Figure 1: Dose–response re-weighting, all partitions. Solid curves are the four decision cells (hardest clean split); dotted are partitions we do not judge on—ID is not an extrapolation test, and parabola's v-OOD/both-OOD nMSE is denominator-degenerate (Appendix E). Only 2 of the 4 decision cells reach parity (uniform both-OOD, parabola r/m-OOD); uniform r/m-OOD and collision both-OOD stay above baseline at every weight, and collision degrades as weight grows. The sweep is single-seed; both parity cells were re-run over three seeds and land at baseline (§4.3 of the main paper).

without displacing the black box's redundant copy.

**Encoded quantity vs. domain (position monotone, velocity sign-flipping).** The reweighted slot's effect is systematic in what it encodes (hardest clean split; baseline: uniform both-OOD $0.131$, parabola r/m-OOD $0.127$, collision both-OOD $0.393$):

| slot content | uniform | parabola | collision |
|---|---|---|---|
| position (structpos) | 0.132 | 0.160 | 0.596 |
| position+velocity | 0.207 | 0.096 | 0.621 |
| $\Delta$ (adding velocity) | $+0.075$ | $-0.064$ | $+0.025$ |

Position is an output variable with the same role in every domain, so pinning it degrades monotonically with dynamics complexity and never flips sign. The marginal effect of adding velocity ($+0.075/-0.064/+0.025$) tracks velocity's role in each domain—redundant constant (uniform), linear driver (parabola), discontinuous jump (collision). The parabola gain ($0.122 \pm 0.007 \rightarrow 0.096 \pm 0.007$, every seed lower) is thus the sole seed-robust structural gain in the grid: a fixed form landing on a matching domain, not a portable prior.

**Physion++ per-scenario (all ten scenarios, FR at 20 epochs).** Rollout nMSE: bouncy_platform 0.041, bouncy_wall 0.094, friction_cloth 0.071, friction_collision 0.062, friction_platform 0.041, mass_collision 0.083, mass_dominoes 0.058, mass_waterpush 0.122; deformable scenes are the common hard case for every method (deform_clothhang 1.375, deform_clothhit 0.772). Injection variants degrade the rigid-body scenarios by 3–10$\times$ (Table 2) while leaving the deformable failure in place.

**Physion++ horizons.** Overall nMSE at horizons 16/32/64: baseline ($H$=8) 0.016/0.140/0.280; $H$=20 0.022/0.033/0.220; $H$=16+scale 0.010/0.035/0.136; $H$=20+scale 0.012/0.024/0.087; $H$=28+scale reaches

| Scenario | FR | +slot | +cons. | +cons.+acc. |
|---|---|---|---|---|
| friction_platform | **0.041** | 0.119 | 0.078 | 0.067 |
| mass_collision | **0.083** | 0.510 | 0.682 | 0.696 |
| mass_dominoes | **0.058** | 0.452 | 0.586 | 0.597 |
| mass_waterpush | **0.122** | 0.363 | 0.663 | 1.476 |

Table 2: Physion++ direct training: per-scenario rollout nMSE (↓) at 20 epochs. Injection degrades realistic-simulation prediction by 3–10$\times$ on the mass/friction scenarios shown (deformable-object scenes are the hardest category for all methods, including the baseline). Single-seed; the gaps far exceed baseline seed noise.

horizon-64 nMSE 0.014. Free rollout is the main long-horizon lever (horizon-32 cosine 0.91 vs. 0.79–0.87 for every physics-augmented variant); geometric scale jitter acts as an amplitude stabilizer for long rollouts: without it, $H$=20 explodes on friction-collision (nMSE 24.6 at cosine 0.99); with it, 0.019.

## C Cross-Backbone Replication: Frozen DINOv2

The cross-backbone variant (§3.5 of the main paper) freezes a DINOv2-small encoder (Oquab et al. 2024)—in the style of DINO-WM (Zhou et al. 2025)—and trains only the projector (as adapter) and the identical predictor stack, losses, and seeds ($\sim$130 runs). Free-rollout baselines (three-seed): uniform both-OOD $0.427\pm0.039$, parabola r/m-OOD $0.225\pm0.019$, collision both-OOD $0.479\pm0.009$; Physion++ horizon-64 $0.258\pm0.005$ vs. teacher forcing $1.030\pm0.028$.

**Teacher forcing vs. free rollout, per domain.** Values behind Figure 4 of the main paper (hardest clean split, latent nMSE ↓, three-seed mean±sample std).

LeWM: uniform $0.300{\pm}0.007 \rightarrow 0.136{\pm}0.008$ ($2.2\times$), parabola $0.443{\pm}0.048 \rightarrow 0.122{\pm}0.007$ ($3.6\times$), collision $1.152{\pm}0.048 \rightarrow 0.479{\pm}0.079$ ($2.4\times$), Physion++ horizon-64 $1.174{\pm}0.041 \rightarrow 0.141{\pm}0.018$ ($8.3\times$). Every teacher-forcing interval is disjoint from its free-rollout counterpart. The frozen-DINOv2 counterparts above give $1.4$–$4.0\times$ over the same four settings, so the lever survives replacing a trainable ViT-tiny with a frozen general-purpose encoder.

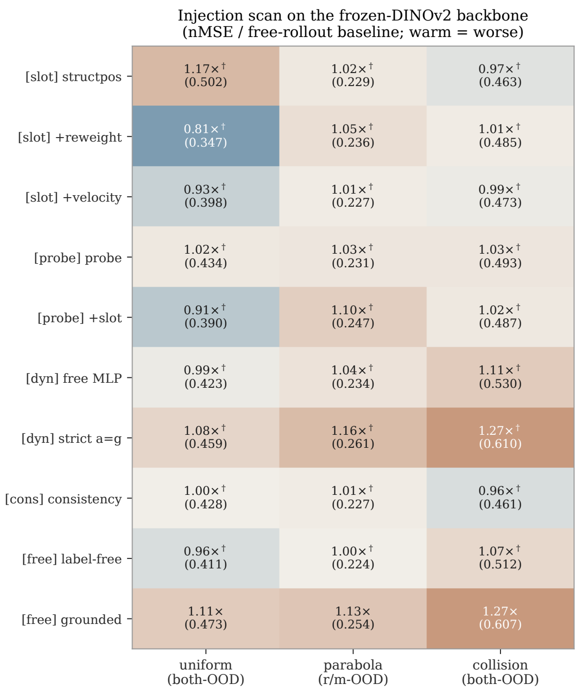


Figure 2: The injection scan repeated on the frozen-DINOv2 backbone (nMSE over the free-rollout baseline; mean over available seeds, † = three seeds). 10 cells worse / 17 parity / 3 better (27/30 at or below baseline). The dynamics-head variants are the worst here too (strict $a{=}g$ and the grounded prior reach $1.27\times$ on collision); the three above-parity cells, all on uniform, are the re-weighting confound of Trap 5.

**Presence and bypass.** The frozen encoder, which has never seen PhyWorld and received no physics supervision or gradient, decodes position from real-frame embeddings at $\rho{=}0.951$ (both-OOD)—*higher* than the PhyWorld-trained LeWM encoder ($0.899$). The $190$ non-slot dimensions alone decode at $0.951$, indistinguishable from the full latent; pinning the slot at weight $300$ leaves the black-box copy at $0.973$. Presence and the bypass are properties of generic visual representations.

**Inert, not harmful.** Unlike the trainable-encoder regime, the core variants (re-weighted slot, probe, consistency) land within the baseline's seed band on all three domains, while the plain slot ($1.42\times$ on uniform) and the dynamics-head variants ($1.15$–$1.29\times$ on all domains) still hurt. With the encoder frozen, injection losses can only reshape the $384{\rightarrow}192$ adapter: the bypass cannot be displaced, which bounds the

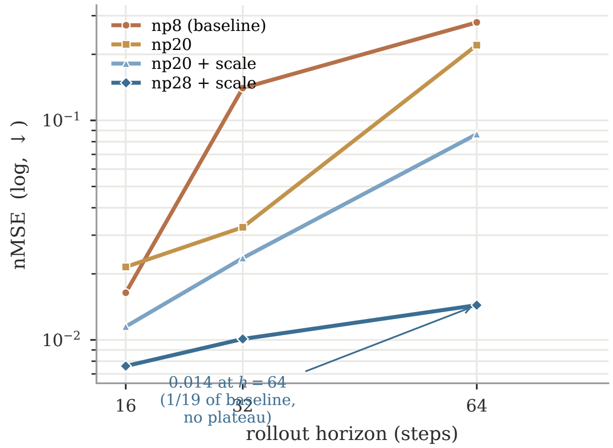


Figure 3: Physion++ direct training: per-horizon rollout nMSE (log scale) as the training horizon grows ($H{=}8{\rightarrow}28$, +scale jitter). Long-horizon error falls monotonically to $1/19$ of baseline with no inflection: realistic-simulation dynamics reward long-rollout training more than the synthetic domains do.

damage—and forecloses the benefit. Injection-variant variance also roughly doubles ($\pm0.087$ vs. baseline $\pm0.039$ on uniform), so single-seed cells on this backbone are uninformative.

# D Augmentation: a Domain-Specific Lever with a Reversal Boundary

Augmentation is not part of the paper's main line (it is neither an injection mechanism nor a protocol variable), but its boundary behavior is a useful caution. On synthetic domains, per-trial appearance jitter (brightness/contrast, strength $0.5$) roughly halves the hardest split: uniform $0.131{\rightarrow}0.068$ ($-48\%$), parabola $-63\%$ (r/m-OOD $0.127{\rightarrow}0.065$); geometric scale jitter, only in combination with a long horizon, sets the collision record ($0.393 \rightarrow 0.172 \pm 0.004$ over three seeds). Interactions are non-monotone: appearance conflicts with long-horizon training ($0.472$, worse than either alone), scale synergizes ($0.208$), the triple is worse than the best pair ($0.253$); temporal-stride augmentation loses even to appearance on velocity-OOD ($0.556$ vs. $0.376$), leaving unseen velocities the open hard case.

The appearance recipe reverses on realistic simulation: the same jitter that halves synthetic OOD error degrades Physion++ friction-collision prediction by two orders of magnitude (nMSE $0.062{\rightarrow}6.44$) while the *aggregate* long-horizon cosine improves (Trap 1 below), and it does not help zero-shot transfer either ($0.597$, below the $0.607$ random-prior ceiling). The boundary is principled: appearance jitter encodes an appearance-invariance assumption that holds under PhyWorld's simple renderer but breaks where appearance carries physical information (friction, mass, material). Scale jitter, which assumes only size-invariance, keeps helping at realistic-simulation scale (Appendix B); appearance jitter flips sign. Augmentation gains are domain- and typespecific, not portable.

## E Metric Pathologies and Evaluation Traps

Our anatomy repeatedly reached conclusions opposite to those suggested by common metrics; the decision rules of §4.1 of the main paper distill the following five failure modes, each of which reversed at least one finding.

**Trap 1: cosine and probe metrics are duals of the training loss.** Probe correlation and latent cosine rise mechanically when probe or slot losses are optimized—they measure whether information is *present*, not whether prediction *uses* it. Probe supervision lifts velocity decodability from $\rho$=0.44 to 0.80 and gives the most stable long-horizon *cosine* (0.882 vs. 0.843), while its *pixel* rollout is the worst in the comparison (19.93 vs. 20.64 dB at horizon 28). Appearance augmentation on Physion++ improves the aggregate long-horizon cosine while per-scene nMSE collapses up to 100× (deform_clothhit: cosine $0.610 \rightarrow 0.913$, nMSE $0.772 \rightarrow 3.69$). Judging either result by direction-only or readout metrics inverts the conclusion. Our own early sweep was misled this way: judged on $K{=}4$ probe $\rho$, $\lambda{=}50$ "won"; re-judged on prediction loss, the ranking reversed.

**Trap 2: nMSE denominators can degenerate.** When the target sequence degenerates (a parabola ball exits the frame, target variance $\rightarrow 0$), per-horizon nMSE explodes—$3\times10^4$–$2\times10^6$ at horizon $\sim$28 across all six parabola baseline variants, while cosine sits at a healthy 0.55–0.95—and pollutes every aggregate that includes late horizons. This is why parabola is judged on its clean r/m-OOD partition throughout. Traps 1 and 2 point in opposite directions (cosine under-reports failure; degenerate nMSE over-reports it); neither metric family is safe alone, hence per-partition, per-horizon cross-validation.

**Trap 3: zero-shot transfer is bounded by the architecture prior.** On Physion OCP, a *randomly initialized* encoder scores mean AUC 0.607. No synthetic-training configuration exceeds it: free rollout approaches it (0.603), physics variants fall below it (0.551–0.582), appearance augmentation 0.579–0.597; training longer approaches the ceiling monotonically (0.554 at epoch 1 to 0.603 at epoch 20) without crossing. Per-scenario, only Support (+0.10) and Collide (+0.07) show signal above the random control. Zero-shot claims must be calibrated against the random-prior floor, or they measure the architecture, not the learning.

**Trap 4: protocol confounds fabricate results in both directions.** Three probe-protocol choices (random vs. pretrained initialization, single-frame vs. stacked probes, probing through the projector) multiply into a false negative: the naive stack reads uniform velocity decodability at $R^2$=0.166, the corrected protocol 0.939. A silent initialization bug (only 24 of 216 loadable encoder keys actually loaded) invalidated an entire 45-configuration sweep before a loading guard caught it. And converged training loss proves nothing about rollout: our clean from-scratch baseline converges to the pretrained model's prediction loss (0.006 vs. 0.005) while its rollout OOD error is 2.8× worse.

**Trap 5: loss re-weighting masquerades as physics.** On the frozen-DINOv2 backbone, heavily re-weighting the pinned position slot (weight 300; the 2 slot dimensions then carry 76% of the prediction loss) improves uniform both-OOD from $0.427\pm0.039$ to $0.286\pm0.015$—apparently the one clear injection win on either backbone. Two content-free controls dissolve it: pinning the slot to a *shuffled*, physically meaningless target at the same weight reaches $0.330\pm0.013$, and weakening the pin 10× while keeping the weighting reaches $0.329\pm0.084$—most of the gain, with no usable physics. All three variants degrade in-distribution error ($0.099 \rightarrow 0.115$–$0.136$), the signature of capacity restriction (the predictor is effectively reduced to a 2-dimensional task), and the same recipe *worsens* collision (1.11×) and parabola (1.14×). An apparent injection gain must be checked against a content-free control at matched loss weighting.

**Auxiliary-loss dominance (measurement behind §4.3 of the main paper, Test 2).** At probe weight $\lambda{=}10$, the weighted auxiliary-to-prediction loss ratios—used as a proxy for gradient magnitude; we do not measure gradient norms directly—are 15× (collision), 44× (uniform), and 125× (parabola); the encoder's effective (participation-ratio) dimensionality collapses by 39–90% relative to baseline (uniform $41 \rightarrow 4$). At $\lambda{=}1$ the ratios are benign, which is the setting used in §4.2 of the main paper.

**Checklist.** (i) Report per-partition *and* per-horizon; never aggregate across horizons without inspecting them. (ii) Judge with scale-aware prediction metrics (nMSE, decoded-pixel PSNR); use cosine and probe $\rho$ as diagnostics only. (iii) Audit nMSE denominators for degenerate targets. (iv) Calibrate transfer against a random-init control, and validate probe protocols on a random encoder.

## References


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